

Three-Dimensional Parameter Expansion Study

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2026-05-06 18:30

Purpose

This side experiment asks which single additional parameter is most useful when added to the current two-dimensional walk-forward search over (k, δ) . The full Layer 2 design vector is

$$\theta = (k, \delta, q, \alpha, \beta, L, a, z_{\text{trend}}, \rho).$$

The baseline optimizes only (k, δ) . Each expansion optimizes (k, δ, x) for one additional parameter x , except for the ρ experiment where (k_0, δ_0, ρ) is optimized because ρ affects online smoothing rather than a static single-fold rule. The report contains two protocols: stitched walk-forward out-of-sample folds and GUI-style continuous online runs.

Dataset, Protocol, and Search Grids

Dataset and Protocol

Data source	Yahoo adjusted SPY
Symbol	SPY
Lookback	5 years
Bar interval	1d
Bars	1255
Train size	252 bars
Test size	63 bars
Cost per side	0.0 bps
Protocol	Stitched OOS folds
Objective	$R^m + D^m + 0.15G$

Search Grids

Parameter	Grid
k	0.8,1.1,1.4,1.8,2.2
δ	1,2,3,5,8
q	[0.0, 5e-08, 1e-07, 2e-07, 5e-07]
α	[0.03, 0.06, 0.1, 0.15]
β	[0.8, 0.88, 0.9, 0.94]
L	[30, 60, 90, 120]
a	[0.02, 0.04, 0.06, 0.08]
z_{trend}	[0.0, 0.25, 0.35, 0.5, 0.75]
ρ	[0.1, 0.25, 0.4, 0.6]

Fixed Initial / Design Parameters

Quantity	Value used unless explicitly optimized
Initial k_0	1.2 in GUI online mode; grid-selected in this stitched OOS report
Initial δ_0	3 in GUI online mode; grid-selected in this stitched OOS report
q	1.0e-07
α	0.06
β	0.90
L	60 bars
a	0.04 log-price units
z_{trend}	0.35
ρ	0.25; only active in the online ρ experiment

Protocol note. The stitched rows are formal out-of-sample folds. A training window is used to choose parameters, the following test window is evaluated, and only test-window wealth curves are stitched together. The GUI-style rows instead run one continuous online-adaptive account over the full dataset, with $(k_0, \delta_0) = (1.2, 3)$ and 5-bar parameter updates. Therefore a GUI Real Data run should be compared with the GUI-style table, not with the stitched baseline.

Initialization and API data note. Unless a row explicitly optimizes a parameter, the GUI-style continuous runs start from $(k_0, \delta_0) = (1.2, 3)$ with $q = 10^{-7}$, $\alpha = 0.06$, $\beta = 0.90$, $L = 60$, $a = 0.04$, $z_{\text{trend}} = 0.35$, and $\rho = 0.25$. The Yahoo API dataset is a rolling market-data snapshot fetched as of May 6, 2026, so rerunning the report on a later date may shift the bar count and the reported returns.

Stitched Walk-Forward Results

Experiment	Optimized	MtM	BH	Gap	Max DD	Sharpe	Calmar	Trades	Avg hold	Δ MtM	Final selected
baseline	k, delta	64.02%	77.29%	-7.48%	-18.76%	0.86	0.75	5	33.20	0.00%	k=1.1, delta=5
q	k, delta, q	67.62%	77.29%	-5.45%	-18.76%	0.89	0.79	7	31.14	3.59%	k=0.8, delta=2, q=5.0e-07
alpha	k, delta, alpha	79.48%	77.29%	1.24%	-18.76%	1.02	0.90	6	31.50	15.46%	k=1.1, delta=2, alpha=0.1
beta	k, delta, beta	64.77%	77.29%	-7.06%	-18.76%	0.86	0.76	5	32.20	0.74%	k=1.1, delta=2, beta=0.94
L	k, delta, L	64.01%	77.29%	-7.49%	-18.76%	0.86	0.75	5	35.40	-0.02%	k=0.8, delta=1, L=30
a	k, delta, a	51.42%	77.29%	-14.59%	-19.48%	0.74	0.60	8	31.38	-12.61%	k=1.1, delta=5, a=0.02
z_trend	k, delta, z_trend	76.53%	77.29%	-0.42%	-18.76%	0.99	0.87	8	25.00	12.51%	k=1.1, delta=5, z_trend=0
rho	k_init, delta_init, rho	70.07%	77.29%	-4.07%	-18.76%	0.95	0.81	9	26.33	6.05%	k_init=0.8, delta_init=2, rho=0.1

GUI-Style Continuous Online Results

Experiment	Swept	Real.	MtM	BH	Gap	Max DD	Sharpe	Calmar	Trades	Avg hold	Δ MtM	Selected + end live
baseline	fixed GUI defaults	42.05%	53.40%	86.13%	-17.59%	-25.56%	0.61	0.35	8	122.88	0.00%	k0=1.2, delta0=3, end k=0.8, end d=2
q	q	80.98%	99.73%	86.13%	7.30%	-17.68%	0.94	0.84	14	72.07	46.33%	k0=1.2, delta0=3, q=0, end k=0.8, end d=2
alpha	alpha	71.88%	89.69%	86.13%	1.91%	-19.50%	0.86	0.70	13	81.00	36.29%	k0=1.2, delta0=3, alpha=0.1, end k=0.8, end d=2
beta	beta	64.69%	77.37%	86.13%	-4.71%	-19.50%	0.79	0.63	12	80.83	23.97%	k0=1.2, delta0=3, beta=0.94, end k=0.8, end d=2
L	L	60.89%	66.01%	86.13%	-10.81%	-25.56%	0.70	0.42	9	131.00	12.61%	k0=1.2, delta0=3, L=120, end k=0.8, end d=2
a	a	77.09%	89.88%	86.13%	2.01%	-22.97%	0.88	0.60	20	49.75	36.48%	k0=1.2, delta0=3, a=0.02, end k=0.8, end d=2
z_trend	z_trend	55.84%	66.89%	86.13%	-10.34%	-19.50%	0.75	0.56	11	84.27	13.49%	k0=1.2, delta0=3, z_trend=0.5, end k=0.8, end d=2
rho	rho	99.72%	115.68%	86.13%	15.88%	-19.50%	1.09	0.86	15	61.93	62.28%	k0=1.2, delta0=3, rho=0.1, end k=0.8, end d=3

Reading the Tables

For the stitched table, MtM is the stitched out-of-sample mark-to-market return. BH is the buy-and-hold return on the same stitched test windows. Gap is $W_T^m / W_T^{bh} - 1$. Max DD is the maximum mark-to-market drawdown. Avg hold is measured in bars. For the GUI-style table, MtM is the full continuous online mark-to-market return over the selected dataset. Δ MtM is measured relative to the baseline row within the same table.

High-Level Takeaways

- Stitched best MtM return: alpha with 79.48%.
- Stitched best benchmark gap: alpha with 1.24%.
- Stitched best Sharpe ratio: alpha with 1.02.
- GUI-style best MtM return: rho with 115.68%.
- GUI-style best benchmark gap: rho with 15.88%.
- GUI-style best Sharpe ratio: rho with 1.09.

Comparison Dataset: NVDA 1-Year Daily

This section repeats the same experiment structure on NVDA using 1 year of 1d bars. Because this sample is shorter than the SPY 5-year dataset, the stitched walk-forward folds use a shorter training window of 126 bars and a test window of 21 bars.

Data source	Yahoo adjusted NVDA
Symbol	NVDA
Lookback	1 year
Bar interval	1d
Bars	251
Train size	126 bars
Test size	21 bars

NVDA 1-Year Daily: Stitched Walk-Forward

Experiment	Optimized	MtM	BH	Gap	Max DD	Sharpe	Calmar	Trades	Avg hold	Δ MtM	Final selected
baseline	k, delta	-14.91%	-6.72%	-8.78%	-24.30%	-0.99	-1.32	0	–	0.00%	k=1.1, delta=1
q	k, delta, q	-14.91%	-6.72%	-8.78%	-24.30%	-0.99	-1.32	0	–	0.00%	k=0.8, delta=1, q=5.0e-07
alpha	k, delta, alpha	-14.91%	-6.72%	-8.78%	-24.30%	-0.99	-1.32	0	–	0.00%	k=1.1, delta=1, alpha=0.03
beta	k, delta, beta	-14.91%	-6.72%	-8.78%	-24.30%	-0.99	-1.32	0	–	0.00%	k=1.1, delta=1, beta=0.8
L	k, delta, L	-14.91%	-6.72%	-8.78%	-24.30%	-0.99	-1.32	0	–	0.00%	k=1.1, delta=1, L=30
a	k, delta, a	-14.91%	-6.72%	-8.78%	-24.30%	-0.99	-1.32	0	–	0.00%	k=0.8, delta=1, a=0.02
z_trend	k, delta, z_trend	-14.91%	-6.72%	-8.78%	-24.30%	-0.99	-1.32	0	–	0.00%	k=1.4, delta=2, z_trend=0
rho	k_init, delta_init, rho	-15.77%	-6.72%	-9.70%	-25.06%	-1.07	-1.35	1	13.00	-0.86%	k_init=0.8, delta_init=1, rho=0.1

NVDA 1-Year Daily: GUI-Style Continuous Online

Experiment	Swept	Real.	MtM	BH	Gap	Max DD	Sharpe	Calmar	Trades	Avg hold	Δ MtM	Selected + end live
baseline	fixed GUI defaults	65.55%	65.55%	77.58%	-6.78%	-21.84%	1.71	3.02	3	75.67	0.00%	k0=1.2, delta0=3, end k=0.8, end d=2
q	q	65.55%	65.55%	77.58%	-6.78%	-21.84%	1.71	3.02	3	75.67	0.00%	k0=1.2, delta0=3, q=0, end k=0.8, end d=2
alpha	alpha	65.55%	65.55%	77.58%	-6.78%	-21.84%	1.71	3.02	3	75.67	0.00%	k0=1.2, delta0=3, alpha=0.03, end k=0.8, end d=2
beta	beta	65.55%	65.55%	77.58%	-6.78%	-21.84%	1.71	3.02	3	75.67	0.00%	k0=1.2, delta0=3, beta=0.8, end k=0.8, end d=2
L	L	66.68%	66.68%	77.58%	-6.14%	-21.84%	1.76	3.07	3	74.00	1.13%	k0=1.2, delta0=3, L=30, end k=0.8, end d=2
a	a	65.55%	65.55%	77.58%	-6.78%	-21.84%	1.71	3.02	3	75.67	0.00%	k0=1.2, delta0=3, a=0.04, end k=0.8, end d=2
z_trend	z_trend	68.64%	68.64%	77.58%	-5.03%	-16.56%	1.84	4.17	5	43.00	3.09%	k0=1.2, delta0=3, z_trend=0, end k=0.8, end d=2
rho	rho	100.15%	100.15%	77.58%	12.71%	-10.72%	2.31	9.40	8	27.38	34.60%	k0=1.2, delta0=3, rho=0.6, end k=0.8, end d=1

NVDA 1-Year Daily takeaways. Stitched best MtM is baseline with -14.91%; stitched best Sharpe is baseline with -0.99. GUI-style best MtM is rho with 100.15%; GUI-style best Sharpe is rho with 2.31.

Caveats

This report is exploratory. It changes one extra parameter at a time and does not prove that a parameter is universally useful. Larger grids and more assets increase the risk of overfitting, so results should be checked across multiple symbols, periods, and bar intervals before being promoted into the main paper. The GUI-style parameter sweeps are full-sample sensitivity checks, not strict out-of-sample evidence.